Introduction to Quantum Field Theory, Lecture notes

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Lesson 1, 23/09

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Lesson 2, 24/09

1 Natural units

From this point onwards, natural units will be used:

$$\hbar = c = 1$$

Note: this messes up dimensionality, but, conveniently, forces mass and energy to have the same units

$$E = mc^2 \underset{c=1}{\rightarrow} E = m,$$

and length and time to have the same dimensionality

$$x = ct \underset{c=1}{\rightarrow} [x] = [t].$$

To convert back into more familiar units the constants are, respectively,

$$c = 299.792.458 \text{ m/s}$$

$$\hbar c \approx 0.97 \text{ GeV·fm}$$

2 Klein-Gordon Equation

The relativistic quantum mechanics' fundamental equation is

$$(\Box + m^2)\psi(x) = 0$$

where

$$\Box = \partial^\mu \partial_\mu = \frac{\partial^2}{\partial t^2} - \nabla^2$$

and $\psi(x)$ is still the Schrödinger's wavefunction.

It will now be shown that this equation **does not** conserve probability, using a continuity equation.

Regular quantum mechanics states

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{j} = 0,$$

where

$$\rho = \langle \psi_t | \psi_t \rangle \qquad \mathbf{j} = \frac{i}{m} (\psi^* \boldsymbol{\nabla} \psi - \psi \boldsymbol{\nabla} \psi^*)$$

Consider the Klein-Gordon equation left-multiplied by $\psi^*(x)$ and its conjugate

$$\begin{cases} \psi^*(x) \left(\Box + m^2\right) \psi(x) = 0 \\ \psi(x) \left(\Box + m^2\right) \psi^*(x) = 0 \end{cases}$$

their difference is

$$\psi^*(\Box + m^2)\psi - \psi(\Box + m^2)\psi^* = 0$$

$$\partial^\mu(\psi^*\partial_\mu\psi - \psi\partial_\mu\psi^*) = 0$$
$$\partial^\mu J_\mu = 0$$

this is a 4D continuity equation and J^μ is a Noether current.

Expanding the equation yields

$$\frac{\partial}{\partial t}\left(\psi^*\frac{\partial\psi}{\partial t} - \psi\frac{\partial\psi^*}{\partial t}\right) - \boldsymbol{\nabla}\cdot(\psi^*\boldsymbol{\nabla}\psi - \psi\boldsymbol{\nabla}\psi^*) = 0$$

the second term is proportional to the quantum probability current, but it is missing an i! Moreover, the first term, representing ρ, is **not definite positive**, so it cannot represent the norm of the state $|\psi_t\rangle$!

This proves that it is **impossible** to make QM relativistic using wavefunctions and retaining the probabilistic approach.

Paul Dirac used another approach, he wanted to decrease the order of space derivative (instead of increasing the time derivative order, like K-G): this result was better, but for free particles it yielded a double solution with $E < 0$ states, violating a fundamental QM theorem!
For this reason, he devised the concept of the *Dirac sea*: these $E < 0$ states must all be occupied (because they cannot be observed), and a uniform charge can be factored out of Maxwell's equations with proper boundary conditions; when one of these particles in the sea is excited, the "hole" left behind can be perceived as a "positive electron": he predicted the existence of *antiparticles*.

This interpretation is still classical, but it begins to convey the concept that a finite number of particles is *not* possible for relativistic QM, so the concept of wavefunctions must be left behind in favour of field theories.

3 Symmetries and conservations

To build a theory, it is necessary to start with principles and symmetry: **Lorentz-invariance** to guarantee the conservation of distance between points (this is also useful to define a metric topology!).

Why symmetry? Noether theorem states that for every 4-current that respects a continuity equation

$$\partial_\mu J^\mu = 0$$

there is a conserved Noether charge

$$Q := \int \mathrm{d}^3x J^0, \text{ so that } \frac{\mathrm{d}Q}{\mathrm{d}t} = 0$$

3.1 Lorentz-Poincaré group

The Lorentz invariant is

$$s^2 = t^2 - x_i x^i = t'^2 - x'_i x'^i = x_\mu x^\mu :$$

the Lorentz group is the set of all transformations $x^\mu \to x'^\mu$ over the Minkowski space that conserve s^2; in matrix representation the elements are

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu$$

the invariant is conserved **if and only if**

$$\boxed{g_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma = g_{\rho\sigma}.}$$

The covariant quantities transform with the inverse matrix

$$x'_\mu = \Lambda_\nu{}^\mu x'_\nu$$

Observation 1 Since $g_{\mu\nu}$ is a symmetric matrix, this equation encompasses 10 constraints on $\Lambda^{\mu}{}_{\nu}$, leaving 3 degrees of freedom for rotations and 3 for boosts.

Naming convention Consider the determinant of the invariance condition:

$$\det\left(\Lambda^T g \Lambda\right) = \det(g) \leftrightarrow \det(\Lambda)^2 = 1 \leftrightarrow \det(\Lambda) = \pm 1$$

Transformations with $\det(\Lambda) = 1$ are called *proper Lorentz transformations.*
Transformations with $\det(\Lambda) = -1$ are called *improper Lorentz transformations.*

Naming convention Consider the (0,0) component of the invariance condition:

$$g_{\mu\nu}\Lambda^{\mu}{}_{0}\Lambda^{\nu}{}_{0} = 1 \leftrightarrow (\Lambda_{00})^2 - \Lambda^{i}{}_{0} g_{ii} \Lambda^{i}{}_{0} = 1 \leftrightarrow |\Lambda^{0}{}_{0}|^2 \geq 1$$

Because this term regulates how the transformation relates x^0 and x'^0, transformations with $\Lambda^{0}{}_{0} \geq 1$ are called *orthochronous Lorentz transformations*, while those with $\Lambda^{0}{}_{0} \leq 1$ are called *non-orthochronous Lorentz transformations.*

These 2 divisions split the Lorentz group into 4 parts

- Proper and orthochronous $L_{+}^{\uparrow}$: restricted Lorentz group, sometimes referred to as "Lorentz Group";
- Improper and orthochronous $L_{-}^{\uparrow}$: spacial inversion;
- Proper and non-orthochronous $L_{+}^{\downarrow}$: time inversion;
- Improper and non-orthochronous $L_{-}^{\downarrow}$: total inversion of spacetime.

Note The only subgroup is $L_{+}^{\uparrow}$, as it is the only one that contains the identity matrix $\mathbb{I}$. **Lesson 3, 28/09**

4 Symmetries and Group Theory

The principle of symmetry (invariance) is the guiding light to create a theory: the first and fundamental symmetry is *Lorentz invariance*, because special relativity states that **all physical laws must be the same in every inertial reference frame**; each particle (state) must also observe other symmetries, but those vary case by case.

The language of symmetries is *group theory*, so, in order to proceed, an introduction to groups is needed.

Def. A group is a set G with a binary operation $\cdot$ so that

- $\forall g_1, g_2 \in G$, $g_1 \cdot g_2 \in G$;
- $\exists e \in G$ so that $\forall g \in G$, $g \cdot e = e \cdot g = g$;
- $\forall g \in G$, $\exists g^{-1}$, so that $g^{-1} \cdot g = g \cdot g^{-1} = e$;
- $\forall g_1, g_2, g_3 \in G$, $g_1 \cdot (g_2 \cdot g_3) = (g_1 \cdot g_2) \cdot g_3 = g_1 \cdot g_2 \cdot g_3$.

Def. A group is *Abelian* if and only if

$$\forall g_1, g_2 \in G,\ g_1 \cdot g_2 = g_2 \cdot g_1$$

4.1 Zoology

- U meaning "unitary" means $UU^{\dagger} = \mathbb{I}$, for complex valued matrices;
- O meaning "orthogonal" means $OO^T = \mathbb{I}$, for real valued matrices
- S meaning "special" means $\det(S) = 1$.

Examples

- $U(1)$ is the group of phase rotations in $\mathbb{C}$;
- $SU(2)$ is the group of spin;
- $SO(3)$ is the group of proper rotations

Note The number in the name of the group is the dimensionality of the fundamental representation.

5 Lie Groups

The most important groups are those that represent *continuous transformations*, i.e. every element of the group corresponds to a transformation that can be broken down into *infinitesimal transformations*: using parameters α and generators T, the element $g(\alpha) \in G$ is

$$g(\alpha) = \mathbb{I} + i\alpha_a T^a + O(\alpha^2) = ... = \exp\left[i\alpha_a T^a\right]$$

so its generators can be found as

$$T^a = -i \left.\frac{\mathrm{d}g}{\mathrm{d}\alpha_a}\right|_{\alpha=0}$$

it is useful to work with generators as their number is small and they are operators (that can be represented as matrices).

The operation of the groups can be used to construct the operation between the generators, by the BCH formula:

$$g_1 \cdot g_2 \to e^X \cdot e^Y := \exp\left(X + Y + \frac{1}{2}[X,Y] + \frac{1}{12}[X,[X,Y]]\right)$$

where $[\cdot,\cdot]$ is the *Lie commutator*.

Def. The vector space of the generators equipped with a commutator (bilinear, anti-commutative, Jacobi Identity) is called a **Lie Algebra** $\mathfrak{g}$.

Note The most general form of a commutator is

$$[T^a, T^b] = i f^a{}_{bc} T^c$$

5.1 Example: algebra of SO(3)

The rotation matrix about the X axis of an angle ϑ_1 is

$$R_1(\vartheta_1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\vartheta_1) & -\sin(\vartheta_1) \\ 0 & \sin(\vartheta_1) & \cos(\vartheta_1) \end{pmatrix}$$

the first generator is then

$$J_1 = -i \left.\frac{\mathrm{d}R_1}{\mathrm{d}\vartheta_1}\right|_{\vartheta_1=0} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}.$$

The other 2 matrices can be obtained similarly:

$$J_2 = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} \qquad J_3 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The commutation relations turn out to be

$$[J^i, J^j] = i\varepsilon_{ijk} J^k.$$

From the generators, the $J^{\pm}$ operators are constructed.

5.2 Example: algebra of SU(2)

The definition states $SU(2) = \{2\text{x}2 \text{ unitary matrices with} \det(U) = 1\}$: the conditions imply, by the BCH formula:

$$U^\dagger U = \mathbb{I} \to J_i^\dagger = J_i$$
$$\det(U) = \det\left(e^{J_i}\right) = 1 = \exp[\mathrm{Tr}(J_i)] \to \mathrm{Tr}(J_i) = 0$$

the traceless, hermitian matrix vector space has base $\sigma^i i$, the Pauli matrices, which have commutation relations

$$[\sigma^i, \sigma^j] = i\varepsilon_{ijk}\sigma^k$$

6 Group representations

Def. A representation of a group G is a map

$$r : G \to GL(V)$$
$$r : g \to \text{square matrix}$$

so that

$$r(g_1 \cdot g_2) = r(g_1) \cdot r(g_2)$$

where the second $\cdot$ is the matrix multiplication.
$\dim(V) = n$ is the dimension of the representation.

Def. An irreducible representation is a representation in which no subset of $\mathrm{Im}(r)$ is a subgroup of $\mathrm{Im}(r)$.

Note A reducible representation can always be block-diagonalized (?). **Lesson 4, 30/09**

The generators are useful because one can study the entire group just by considering the generators: they are operators uniquely defined by their commutation relations using Lie brackets; the set of generators equipped with their commutation relations is the **Lie algebra** associated with the group.

Once the algebra is known, the choice of the representation yields matrices equivalent to the generators.

As first examples, it is useful to consider $SO(3)$ and $SU(2)$, because they are closely related to the Lorentz group (which is the final purpose of this parenthesis on group theory).

7 Example: angular momentum as a representation

As is known, the algebra of the angular momentum operators J^i is

$$\left[J^i, J^j\right] = i\varepsilon^{ij}{}_k J^k$$

the rank of a group is the maximum number of generators that commute with one another: in this case the rank is 1 and the chosen operator is J^3.
A Casimir operator is an operator that commutes with all generators: in this case it is

$$J^2 = J_i J^i.$$

The physical question is then "how many possible values/eigenstate of the angular momentum are possible?" and it turns out that the answer to this question also answers, in a QFT formalism, "how many particles does the field predict?".
The process to find the possible eigenvalues and eigenstates is known (see QM1 course) and it yields the

fact that for each j value, the number of eigenstates of J^3 is finite and their eigenvalues m must be in the range

$$-j < m < j$$

where j is called the *maximum weight*; so for each j the possible values of J^3 (and so the number of different eigenstates) are $2j+1$. From this process also follows that the only possible values of j are positive integers or semi-integers and that the values of m always differ by an integer (so if j is semi-integer, all the corresponding values of m must also be semi-integer and vice-versa).

This formalism is very powerful because it relates the irreducible representations of the algebra to its vector basis: in a QFT, the dimension of the vector space will correspond to the number of possible physical particles.

This algebra is common between $SU(2)$ and $SO(3)$, with the key difference that for $SO(3)$ only integer values of j are permitted: as seen in intro groups, $SU(2)$ can be represented as a 3-sphere immersed in $\mathbb{R}^4$, while $SO(3)$ is only half 3-sphere in which antipodal points are mapped to the equator (??).

8 Example: reducible representation

The combination of 2 spin $\frac{1}{2}$ particles yields a reducible representation: the vector space is

$$V = \mathbb{C}^2 \otimes \mathbb{C}^2, \quad \dim(V) = 4$$

again, the process of spin combination is treated in the QM1 and QM2 courses and it yields a *triplet* and a *singlet* of states:

$$\text{Triplet (J=1):} \begin{cases} |J=1, M=-1\rangle \\ |J=1, M=0\rangle \\ |J=1, M=-1\rangle \end{cases} \rightarrow \text{Vector states}$$

$$\text{Singlet J=0} : |J=0, M=0\rangle \rightarrow \text{Scalar state}$$

So the space V is subdivided into two different vector spaces which differ in symmetry: the singlet is antisymmetric while the triplet is symmetric; this representation is reducible, because the image of the representation of the algebra is the direct sum of the images of the representations of two different algebras.

9 The Lorentz Group Representation

Only the subgroup $L_+^\uparrow$ is here considered and its formal name is $SO(1,3)$, where $(1,3)$ is the signature of the Minkowski metric $(+,-,-,-)$.

Consider an infinitesimal Lorentz transformation

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \quad |\omega^\mu{}_\nu| << 1 \, \forall \mu, \nu$$

where $\omega^\mu{}_\nu$ are called *parameters* of the transformation.

Substituting into the invariance condition yields

$$g_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma = g_{\rho\sigma}$$
$$g_{\mu\nu}\left(\delta^\mu{}_\rho + \omega^\mu{}_\rho\right)\left(\delta^\nu{}_\sigma + \omega^\nu{}_\sigma\right) = g_{\rho\sigma}$$
$$g_{\mu\nu}\delta^\mu{}_\rho\delta^\nu{}_\sigma + \omega_{\nu\rho}\delta^\nu{}_\sigma + \omega_{\mu\sigma}\delta^\mu{}_\rho = g_{\rho\sigma}$$
$$\omega_{\rho\sigma} = -\omega_{\rho\sigma}$$

so the parameters must be **antisymmetric**. This confirms the fact that each transformation is uniquely determined by 6 quantities. The transformation becomes

$$\Lambda^{\mu}{}_{\nu} = \delta^{\mu}{}_{\nu} + g^{\mu\rho} g^{\sigma}{}_{\nu} \omega_{\rho\sigma}$$

since $\omega_{\rho\sigma}$ are antisymmetric, the only non-zero part of the product $\omega_{\rho\sigma} g^{\mu\rho} g^{\sigma}{}_{\nu}$ is given by the antisymmetric part of $g^{\mu\rho} g^{\sigma}{}_{\nu}$: anti-symmetrizing it yields

$$\Lambda^{\mu}{}_{\nu} = \delta^{\mu}{}_{\nu} + \frac{1}{2} \omega_{\rho\sigma} \left(g^{\mu\rho} g^{\sigma}{}_{\nu} - g^{\mu\sigma} g^{\rho}{}_{\nu} \right) = \delta^{\mu}{}_{\nu} + i \frac{1}{2} \omega_{\rho\sigma} \left(M^{\rho\sigma} \right)^{\mu}{}_{\nu}$$

confronting with the definition of generators

$$g = \mathbb{I} + i \alpha_a T^a$$

yields that the generators are

$$M^{\rho\sigma} = -i \left(g^{\mu\rho} g^{\sigma}{}_{\nu} - g^{\mu\sigma} g^{\rho}{}_{\nu} \right)$$

so the generic Lorentz transformation can be expressed as

$$\Lambda = \exp \left[\frac{i}{2} \omega_{\rho\sigma} M^{\rho\sigma} \right]$$

and its commutation relations are

$$\left[M^{\lambda\tau}, M^{\rho\sigma} \right] = -i \left(g^{\lambda\sigma} M^{\tau\rho} + g^{\tau\rho} M^{\lambda\sigma} - g^{\lambda\rho} M^{\tau\sigma} - g^{\tau\sigma} M^{\lambda\rho} \right)$$

To make computations easier, the 6 generators are split into 3+3 easier operators: since $SO(3)$ is a subgroup of $SO(1,3)$, there is a way to express half of the Lorentz Group generators in terms of those of $SO(3)$:

$$J_i = \frac{1}{2} \varepsilon_{ijk} M^{jk}$$

which obey the $SO(3)$ algebra

$$[J_i, J_j] = i \varepsilon_{ij}{}^{k} J_k$$

and generate rotations; the boosts are generated by

$$K^i = M^{0i}$$

which obey the commutation relation

$$\left[K^i, K^j \right] = -i \varepsilon^{ij}{}_{k} J^k$$

which means that the algebra of the K's is **not** closed, whereas the algebra of the J's is: this is due to the fact that 3D rotations form a subgroup of $SO(1,3)$, while boosts do not (it is easy to check that the composition of 2 Cartesian boosts is the composition of a Cartesian boost and a rotation). The mixed commutator yields

$$[J_i, K_j] = i \varepsilon_{ijk} J^k$$

so the set of all J's and all K's is the set of the generators of $SO(1,3)$.

Lesson 5, 01/10

10 More on the Lorentz group

The generic Lorentz transformation can also be written in terms of J, K:

$$\Lambda = \exp\left[i\vartheta_i J^i + i\eta_i K^i\right]$$

where ϑ_i are the Euler angles (parameters of the rotation) and η_i are the rapidities (parameters of the boost).

With some work, it is possible to prove that

$$\frac{1}{2}M_{\mu\nu}M^{\mu\nu} = J_i J^i - K_i K^i = \mathbf{J^2} - \mathbf{K^2}$$

and

$$-\frac{1}{4}\varepsilon^{\mu\nu\rho\sigma}M_{\mu\nu}M_{\rho\sigma} = J_i K^i = \mathbf{J}\cdot\mathbf{K}$$

are Casimir operators for the representation: there exist only 2 linear independent Casimir operators and each and any linear combination of these two is valid, but these two are important because they are, respectively, symmetric and antisymmetric (how can they be?? They're scalar!).

Theorem: The algebra of the Lorentz Group can be factorized as the direct sum of two $SU(2)$ algebras

$$\mathfrak{so}(1,3) = \mathfrak{su}(2) \oplus \mathfrak{su}(2).$$

This suggests to find new generators that are suited to this decomposition

$$\mathbf{N} = \frac{1}{2}\left(\mathbf{J} + i\mathbf{K}\right)$$
$$\mathbf{N}^\dagger = \frac{1}{2}\left(\mathbf{J} - i\mathbf{K}\right)$$

their algebra is

$$\left[N_i, N_j^\dagger\right] = 0$$
$$[N_i, N_j] = i\varepsilon_{ijk}N^k$$
$$\left[N_i^\dagger, N_j^\dagger\right] = i\varepsilon_{ijk}N^{k\dagger}$$

Observation: These generators are convenient, since the algebra of the N's and that of the $N^\dagger$'s are closed and disjoint.

The Casimir operators are

$$N^2 = N_i N^i \longrightarrow \underset{\text{eigenvalues}}{n(n+1)}$$
$$N^{\dagger 2} = N_i^\dagger N^{i\dagger} \longrightarrow m(m+1),$$

so the irreducible representations of the group will be labeled by the pair (n, m). Since they obey $SU(2)$ algebra, it is clear that n, m will have integer or semi-integer values and that the eigenvalues of $N_3, N_3^\dagger$, s and $s^\dagger$, will respect

$$-n < s < n \qquad -m < s^\dagger < m.$$

(n,m)	Field name	Dim	Example
$(0,0)$	Scalar φ	1	Higgs
$(1/2,0)$	Spinor ψ_L	2	Weyl Spinor
$(0,1/2)$	Spinor ψ_R	2	Weyl Spinor
$(1/2,0)\oplus(0,1/2)$	Dirac Spinor	4	
$(1/2,1/2)$	4-vector field (spin 1)	4	A^μ EM
$(1,0)\oplus(0,1)$	Rank 2 antisymm tensor	6	$F^{\mu\nu}$ EM

To each pair n,m corresponds a name for the field

The generators for each field can be derived: for scalar particles it is easy, consider

$$n=0\rightarrow \mathbf{N}=0=\frac{1}{2}\left(\mathbf{J}+i\mathbf{K}\right)$$
$$m=0\rightarrow \mathbf{N}^\dagger=0=\frac{1}{2}\left(\mathbf{J}-i\mathbf{K}\right)$$

which has only one solution.

$$\mathbf{J}=\mathbf{K}=0$$

For the spinor $(1/2,0)$,

$$n=1/2\rightarrow \mathbf{N}=\frac{1}{2}\sigma$$
$$m=0\rightarrow \mathbf{N}^\dagger=0$$

which implies

$$\begin{cases}(\mathbf{J}+i\mathbf{K})=\sigma\\ \mathbf{J}=i\mathbf{K}\end{cases}\rightarrow\begin{cases}\mathbf{J}=\frac{i}{2}\sigma\\ \mathbf{K}=-\frac{i}{2}\sigma\end{cases}$$

and so the generic transformation can be written as

$$\Lambda_{(1/2,0)}=\exp\left[\frac{i}{2}\left(\vartheta_a\sigma^a-i\eta_a\sigma^a\right)\right].$$

A similar process can be done for $(0,1/2)$ and it yields

$$\Lambda_{(0,1/2)}=\exp\left[\frac{i}{2}\left(\vartheta_a\sigma^a+i\eta_a\sigma^a\right)\right]$$

These two spinorial fields will be united into a single, 4-component field called a *Dirac Field*, (by taking the direct sum of the two components), which has some symmetry properties that these two do not possess singularly and describes **the electron**. Their direct product will yield a vector field that describes **the photon**.

11 Poincaré Group

The QFT must also be invariant over spacetime *translations*: the Poincaré group includes Lorentz transformations *and* translations

$$x^\mu\rightarrow x'^\mu:=\Lambda^\mu{}_\nu x^\nu+a^\mu$$

the group parameters are $6+4=10$. The generators of the 4-translations, similarly to QM, are the 4-momenta

$$P^\mu=(H,P^i);$$

moreover, in QM1 the time evolution operator was found to be

$$U = e^{-iH}$$

which, in hindsight, is an exponential map, in which H is the generator and t is the parameter!

There will be some problems with the fact that the translation group is not compact.

Finding the commutation relations for the algebra is not trivial, since rotations and translations do not commute, even in Minkowski space. Consider two successive Poincaré transformations:

$$x^\mu \to x'^\mu = (\Lambda_{(1)})^\mu{}_\nu x^\nu + a_1^\mu \to x''^\mu = (\Lambda_{(2)})^\mu{}_\rho (\Lambda_{(1)})^\mu{}_\nu x^\nu + (\Lambda_{(2)})^\mu{}_\rho a_1^\rho + a_2^\mu$$

it is clear that they do not commute.
The commutation relations are

$$\begin{aligned}
&[P^\mu, P^\nu] = 0 \\
&[M_{\mu\nu}, P_\rho] = -ig_{\rho\sigma}P_\nu + ig_{\nu\rho}P_\mu
\end{aligned}$$

They are clearer when time and space indexes are expressed separately

$$\begin{aligned}
&\left[P^i, P^j\right] = 0 \quad \left[P^i, H\right] = 0 \quad \left[K^i, P^j\right] = i\delta^{ij}H \\
&\left[J^i, J^j\right] = i\varepsilon^{ij}{}_k J^k \quad \left[K^i, H\right] = iP^i \quad \left[J^i, P^j\right] = i\varepsilon^{ij}{}_k P^k \\
&\left[J^i, H\right] = 0 \quad \left[K^i, K^j\right] = i\varepsilon^{ij}{}_k J^k \quad \left[J^i, K^i\right] = i\varepsilon^{ij}{}_k K^k
\end{aligned}$$

Next, the Casimir operators: the expected number is 2, as there are only 2 commuting generators, a natural choice is

$$P^2 = P_\mu P^\mu \underset{\text{eigenvalue}}{\longrightarrow} m^2,$$

where m is the mass of the particle (as the mass-shell dictates); a not so natural choice is the *Pauli-Lubanski vector* squared:

$$W^\mu = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}P_\nu M_{\rho\sigma} \to W^2 = W_\mu W^\mu.$$

A nice relation is

$$P_\mu W^\mu = 0$$

for antisymmetry reasons, and one can show that

$$[W^\mu, P^\nu] = 0$$

Lesson 6, 05/10

12 Massive and massless particles representations

Massive particles
It is known from SR that if a particle is massive, there exist a reference frame in which the particle is at rest, i.e.

$$P^\mu = (m, \mathbf{0})$$

which means that

$$W^0 = 0 \quad W^i = -\frac{1}{2}\varepsilon^{0ijk}M_{jk}$$

recalling the definition of J^i this is

$$W^i = -\frac{1}{2}mJ^i$$

since the particle is at rest, this cannot represent an orbital angular momentum, so it must be intrinsic, i.e. *spin*:

$$W^i = -\frac{1}{2}mS^i$$

and so massive particles are identified by Casimir operators P^2 and $S^2 \propto W^2$ and by the commuting operators S_3 and P^μ: there are, for each value of m, s, $2s+1$ internal degrees of freedom.

Massless particles
If the particle is massless,

$$P_\mu P^\mu = m^2 = 0$$

which means

$$W_\mu W^\mu = 0,$$

and in general

$$P_\mu W^\mu = 0 :$$

from these 3 conditions it can be shown[2] that

$$W^\mu = hP^\mu$$

where $h = \pm 1$ is called **helicity** and represents the projection of the angular momentum on the direction of the momentum

$$h = \frac{\mathbf{J}\cdot\mathbf{P}}{|\mathbf{P}|};$$

in this case, they can only be parallel or anti-parallel: these states are called *polarizations* and, for photons, they correspond to light polarizations.

This framework also allows a sector of the theory in which spin is continuous ($m = 0$, $W^2 < 0$ states are like this, but they have not been observed in nature) and a sector in which some particles, tachyons can move at $v > c$.

The key takeaway from this arguments is that

The irreducible representations of the Poincaré Group are in one-to-one correspondence with the possible types of elementary particles.
The mathematical structure of space-time symmetries determines what kinds of particles can exist in nature.

13 Non-finite representations of groups: field representations

Until this point, all the representations have been finite-dimensional, with generators represented as matrices, but a field is continuous.

Def. A field is a well-behaved function of spacetime

$$\phi_a(x^\mu) = \begin{pmatrix} \phi_1(x^\mu) \\ \phi_2(x^\mu) \\ \dots \end{pmatrix}$$

matrices can transform a field into another with linear combinations of the old components as new components: these transformations are finite-dimensional, while the ones on the coordinates x^μ can also be infinite-dimensional.

[2] moodle?

In QM the allowed transformations were unitary transformations, as they preserve squared norms, which are the observable part of the theory: these transformations are infinite-dimensional (think of the time evolution: its generator is the Hamiltonian H, which is a differential operator).

In parallel to the definition of the angular momentum operator in QM, the Lorentz generator for infinite-dimensional transformation is

$$M_{\mu\nu} = i(x_\mu \partial_\nu - x_\nu \partial_\mu)$$

and so a field transforms like

$$\phi_a(x) \to \phi'_a(x') = M^a{}_b(\Lambda)\phi_b(\Lambda x)$$

where Λ are infinite-dimensional transformations and $M^a{}_b$ are finite-dimensional; the total transformation can be expressed with the exponential map

$$\phi_a(x') = \exp\left[-\frac{i}{2}\omega^{\rho\sigma} M_{\rho\sigma}\right]\phi_b(x),$$

with

$$M_{\mu\nu} = M^{\text{inf}}_{\mu\nu} + M^{\text{fin}}_{\mu\nu} :$$

where the finite part generates the transformations of the field components[3] and the infinite part those of the space-time coordinates. This generators can be expressed together thanks to the BCH formula.

Lesson 7, 06/10

14 Classical Field theory

Before delving into QFT, it is necessary to understand classical field theory.

A classical system is described by the generalized coordinates

$$q_i(t), \text{ with } i = 1, 2...N$$

where N is the (finite) number of degrees of freedom and t is a parameter. From this the Lagrangian is constructed

$$L(q, \dot{q}).$$

A field theory replaces the set of the coordinates$\{q_i\}$ with a *continuous function of space-time*, called a **field**:

$$\phi(t, \mathbf{x}), \ \mathbf{x} \in \mathbb{R}^3$$

the continuous coordinates $\mathbf{x}$ replace the discrete index i and so the degrees of freedom become infinite.

The next step is to construct a Lagrangian density

$$\mathcal{L}(\phi, \partial_\mu \phi)$$

and solve for the equations of motion.

Example: from finite to infinite degrees of freedom.
Suppose a system of N point-like masses interacting with an elastic potential: the Lagrangian will be

$$L = \sum_{k=1}^{N}\left[\frac{1}{2}m_i\dot{x}_i^2 - \frac{1}{2}k(x_i - x_{i+1})^2\right].$$

[3] Q: why are we assuming that the parameters are the same number and are the same?

Consider $m_i = m$ and the distance between them to be constant and equal to ε:

$$L = \sum_{k=1}^{N} \frac{\varepsilon}{2} \left[\frac{m}{\varepsilon} \dot{x}_i^2 - k \frac{(x_i - x_{i+i})^2}{\varepsilon} \right]$$

The limit as ε goes to 0 is

$$\lim_{\varepsilon \to 0} L = \int \mathrm{d}x \frac{1}{2} \left[\mu \dot{\phi}(x)^2 - Y \left(\frac{\partial \phi}{\partial x} \right)^2 \right]$$

where $\frac{m}{\varepsilon} \to \mu$, $k\varepsilon \to Y$ (which is a Young module) and $x_i(t) \to \phi(t, x)$. Applying the least action principle yields the Euler-Lagrange equation

$$\frac{\partial^2 \phi}{\partial x^2} - \frac{\mu}{Y} \frac{\partial^2 \phi}{\partial t^2} = 0.$$

In general, the Lagrangian density is a function of the spacetime coordinates x^μ through the dependency on a field and its derivatives

$$\mathcal{L}(x^\mu) = \mathcal{L}(\phi, \partial_m u \phi)$$

and the Lagrangian is its integral on the physical space

$$L(t) = \int \mathrm{d}^3 x \mathcal{L}(\phi, \partial_\mu \phi);$$

another important quantity is the action, i.e. the integral over spacetime of the density:

$$S = \int \mathrm{d}^4 x \mathcal{L}(\phi, \partial_\mu \phi).$$

All the theories constructed must be *local*, i.e. they must obey causality, so in the Lagrangian density $\mathcal{L}$ will never present terms like

$$\phi(x^\mu)\phi(y^\mu), \text{ with } x^\mu \neq y^\mu$$

which represents an instantaneous interaction between two points of spacetime.

Def. A field theory is local when the dependency con x^μ is

$$\mathcal{L}(x^\mu) = \mathcal{L}(\phi(x^\mu), \partial_\mu \phi(x^\mu))$$

which means that the dynamics of the system do not connect different points in space instantaneously.

14.1 Transformations of fields

Since the theory must be covariant, it is pivotal to understand how fields behave under Poincaré transformations: consider a point in spacetime expressed in coordinates as x^μ and the value of a field at that point $f(x^\mu)$; consider then an infinitesimal transformation $x'^\mu = x^\mu + \delta x^\mu$ so that

$$\Delta f = f'(x'^\mu) - f(x^\mu) = f'(x^\mu + \delta x^\mu) - f(x^\mu) = f'(x^\mu) - f(x^\mu) + \partial_\mu f'(x^\mu) \delta x^\mu + O(\delta^2 x^\mu)$$

at first order, noting that $\partial_\mu f'(x^\mu) = \partial_\mu f(x^\mu) + O(\delta x^\mu)$, this can be written as

$$\Delta f = \delta f + \partial_\mu f(x^\mu) \delta x^\mu.$$

This can be seen as the definition of the operator Δ on fields

$$\Delta = \delta + \delta x^\mu \partial_\mu :$$

under a Poincaré transformation, the variation of a field is made up of the functional variation in the field itself (δ) and the variation in the field given by the transformation of the coordinates (called transport term).

Example: spacetime translations.
Consider the infinitesimal Poincaré transformation

$$x^\mu \to x'^\mu = x^\mu + \varepsilon^\mu$$

by definition, a local field f is invariant under translations, so

$$\Delta f = 0$$

and this means

$$\delta f = -\varepsilon^\mu \partial_\mu f$$

and since $P_\mu = i\partial_\mu$,

$$\delta f = -i\varepsilon^\mu P_\mu f.$$

This means that this functional transformation of the field is generated by the 4-momenta

$$f' = \exp\left[-i\varepsilon^\mu P_\mu\right] f$$

The Lorentz transformations, on the other hand, depend on the representation the field belongs to: the simplest is the scalar representation.

Example: Lorentz transformation of a scalar field.
A scalar field is a field invariant under Poincaré transformations

$$\phi(x^\mu) = \phi'(x'^\mu).$$

As seen before, the translations are easy

$$\phi(x^\mu) \to \phi'(x'^\mu) = \exp\left[-i\varepsilon^\mu P_\mu\right] \phi(x^\mu).$$

Consider now a generic Lorentz transformation

$$x^\mu \to x'^\mu = x^\mu + \varepsilon^{\mu\rho} x_\rho$$

where $\varepsilon^{\mu\rho}$ are the 6 completely antisymmetric parameters of the transformation.
By definition,

$$\Delta\phi = 0 = \delta\phi + \varepsilon^{\mu\rho} x_\rho \partial_\mu \phi$$

and so

$$\delta\phi = -\varepsilon^{\mu\rho} x_\rho \partial_\mu \phi \to \delta = -\varepsilon^{\mu\rho} x_\rho \partial_\mu$$

this operator is non-zero only for the antisymmetric part of $x_\rho \partial_\mu$, so antisymmetrizing yields

$$\delta = -\varepsilon^{\mu\rho} \frac{1}{2} \left(x_\rho \partial_\mu - x_\mu \partial_\rho\right) = -\varepsilon^{\mu\rho} \frac{i}{2} L_{\rho\mu}$$

where $L_{\rho\mu}$ is the generalized angular momentum expressed in its infinite-dimensional form, related to the $M_{\rho\mu}$ generators.
Observation: no spin operator is found, as expected, because the scalar field belongs to the $(0,0)$ representation, and so the only possible angular momentum of the system is orbital.

And so the general Lorentz transformation on a scalar field is

$$\phi'(x'^\mu) = \exp\left[-\frac{i}{2}\varepsilon^{\mu\rho} L_{\rho\mu}\right] \phi(x^\mu)$$

Observation: Any power of a scalar field is a Lorentz scalar (and so all functions of ϕ expressed as power series), the derivative of ϕ is not scalar, while the double derivative is scalar.

Example: Lorentz transformation of a vector field.
Consider the gradient vector field

$$\partial_\mu \phi,$$

where ϕ is a scalar field: what does the operator Δ do to this kind of field?

$$\Delta \partial_\mu \phi = \delta \partial_\mu \phi + \delta x^\nu \partial_\nu \partial_\mu \phi$$

since $\Delta\phi = 0$, this is also

$$\Delta \partial_\mu \phi = [\Delta, \partial_\mu]\, \phi = [\delta, \partial_\mu]\, \phi + [\delta x^\nu \partial_\nu, \partial_\mu]$$

since δ is a linear operator, it commutes with the 4-derivative; expanding δx^ν yields

$$\Delta \partial_\mu \phi = [\varepsilon^{\nu\rho} x_\rho \partial_\nu, \partial_\mu] = \varepsilon^{\nu\rho} x_\rho \partial_\nu (\partial_\mu \phi) - \varepsilon^{\nu\rho} \partial_\mu (x_\rho \partial_\nu \phi)$$

expanding with the product rule

$$\Delta \partial_\mu \phi = \varepsilon^{\nu\rho} x_\rho \partial_\nu \partial_\mu \phi - \varepsilon^{\nu\rho} x_\rho \partial_\mu \partial_\nu \phi - \varepsilon^{\nu\rho} (\partial_\mu x_\rho) \partial_\nu \phi$$

the first two terms are zero, because $\varepsilon^{\nu\rho}\left(x_\rho \partial_\nu \partial_\mu - x_\rho \partial_\mu \partial_\nu\right) = 0$, and $\partial_\mu x_\rho = g_{\mu\rho}$, so

$$\Delta \partial_\mu \phi = -\varepsilon^{\nu\rho} g_{\mu\rho} \partial_\nu \phi = \varepsilon^{\rho\nu} g_{\mu\rho} \partial_\nu \phi.$$

So finally

$$\begin{aligned} \delta(\partial_\mu \phi) + \varepsilon^{\rho\sigma} x_\rho \partial_\sigma \partial_\mu \phi &= \varepsilon^{\rho\nu} g_{\rho\mu} \partial_\nu \phi \\ \delta(\partial_\mu \phi) &= \varepsilon^{\rho\sigma} g_{\mu\rho} g^\nu{}_\sigma (\partial_\nu \phi) - \varepsilon^{\rho\sigma} x_\rho \partial_\sigma (\partial_\mu \phi) \end{aligned}$$

antisymmetrizing both addends with respect to the indices of ε yields

$$\delta(\partial_\mu \phi) = \frac{1}{2} \varepsilon^{\rho\sigma} \left[\underbrace{\left(g_{\mu\rho} g^\nu{}_\sigma - g_{\mu\sigma} g^\nu{}_\rho\right)}_{\text{Independent of } x^\mu} (\partial_\nu \phi) - \underbrace{\left(x_\rho \partial_\sigma - x_\sigma \partial_\rho\right)}_{x^\mu\text{-dependent}} (\partial_\mu \phi) \right].$$

This form is convenient, because it is clear that the transformation of the field has both a x^μ-dependent generator, which is proportional to the generalized angular momentum $L_{\rho\sigma}$, and a generator that does not depend from x^μ, which corresponds to a spin operator $(S_{\rho\sigma})_\mu{}^\nu$: this spin operator is none other than the generator of the finite-dimensional Lorentz transformations $(M_{\rho\sigma})_\mu{}^\nu$ and the orbital angular momentum operator that of the infinite-dimensional transformations.

$$\delta(\partial_\mu \phi) = -\frac{i}{2} \varepsilon^{\rho\sigma} \left(S_{\rho\sigma}\right)_\mu{}^\nu (\partial_\nu \phi) - \frac{i}{2} \varepsilon^{\rho\sigma} L_{\rho\sigma} (\partial_\mu \phi)$$

This example underlines the fact that vector fields can also present internal degrees of freedom, identified as spin, as they are $(n, m) \neq (0, 0)$ irreducible representations of the Poincaré group. **Lesson 8, 07/10**

15 Constructing a Lagrangian

The guiding principle of this process is the Occam's Razor: nature will realize the simplest Lorentz-invariant terms in the Lagrangian.

The fundamental quantity to build a Lagrangian theory is **the action**, defined before as

$$S = \int \mathrm{d}^4x \mathcal{L}(\phi, \partial_\mu \phi) :$$

in classical mechanics the physical trajectory is the one on which the action (as a functional) is minimal; in QM, even if no trajectory exists, this process can be still used in path integral formulation.

As before stated, the Lagrangian density must be *local*, to guarantee causality, *real*, to preserve probability, *Poincaré invariant*, to ensure that the theory is relativistic and must contain *at most a second derivative*, again to guarantee causality.

Other invariances are admitted (and usually present) and are called *internal symmetries*: for example the invariance of a spinor under a global phase shift guarantees **charge conservation**.

16 Least action

The action in natural units is dimensionless, so the Lagrangian density must have units $[\mathrm{m}^{-4}]$=$[\mathrm{kg}^4]$.

Consider an internal symmetry (i.e. a transformation that does not change the spacetime coordinates)[4]:

$$\phi(x) \to \phi'(x') = \phi(x) + \delta\phi(x),$$

the action

$$S = \int \mathrm{d}^4x \mathcal{L}$$

is minimal if

$$\delta S = 0 = \int \mathrm{d}^4x \delta\mathcal{L} = \int \mathrm{d}^4x \left[\frac{\partial \mathcal{L}}{\partial \phi}\delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu(\delta\phi)\right],$$

where the relation $\delta(\partial_\mu\phi) = \partial_\mu(\delta\phi)$ was used; consider

$$\partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\delta\phi\right) = \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu(\delta\phi) + \partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\right)\delta\phi \to$$
$$\to \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu(\delta\phi) = \partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\delta\phi\right) - \partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\right)\delta\phi :$$

substituting it into the $\delta S = 0$ condition yields

$$0 = \int \mathrm{d}^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\right]\delta\phi + \int \mathrm{d}^4x \partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\delta\phi\right).$$

If the variation of the fields $\delta\phi$ goes to zero at the boundary of the surface enclosing the dominion of integration (i.e. infinity)[5], the second integral is zero by the divergence theorem and the action is stationary if and only if

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial L}{\partial(\partial_\mu\phi)} = 0}$$

which is called the **Euler-Lagrange equation** for fields.

[4] this is equivalent to the small variations of the trajectory in Lagrangian Mechanics.
[5] this is the case for all physical fields.

Note: the E-L equation is invariant under transformations in the form

$$\mathcal{L}' = \mathcal{L} + \partial_\mu \Lambda^\mu$$

where Λ^μ is any 4-vector function: these are called *canonical transformations*. A trivial example of these transformations are simple translations of $\mathcal{L} \to \mathcal{L} + C$.

17 Noether's Theorem

This is one of the pivotal theorems of QFT and all of theoretical physics and it states

To each symmetry of a system corresponds a 4-current satisfying a continuity equation and a conserved quantity, called Noether charge.

Consider a transformation that acts both on the coordinates and a field

$$\begin{cases} x'^\mu = x^\mu + \delta x^\mu \\ \phi'(x') = \phi(x) + \Delta\phi(x) \end{cases}$$

which does not change the action S[6]. Staring with ϕ as a solution to the E-L equation, the action is invariant under the transformation if and only if

$$\Delta S = 0 = \int \mathrm{d}^4x \Delta\mathcal{L} + \int \delta(\mathrm{d}^4x)\mathcal{L}$$

where

$$\Delta\mathcal{L} = \underbrace{\frac{\partial\mathcal{L}}{\partial\phi}\delta\phi + \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\partial_\mu\delta\phi}_{=\delta\mathcal{L}} + \partial_\mu\mathcal{L}\delta x^\mu = \underbrace{\left(\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\right)}_{=0 \text{ E-L}}\delta\phi + \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\delta\phi\right) + (\partial_\mu\mathcal{L})\delta x^\mu.$$

In this case, since the transformation acts on the coordinates, the variation of the integration measure must be considered:

$$\delta(\mathrm{d}^4x) = \mathrm{d}^4x' - \mathrm{d}^4x$$

using the Jacobian of the coordinate transformation

$$\mathrm{d}^4x' = |\det(J(x))|\mathrm{d}^4x = \left|\det\frac{\partial x'^\mu}{\partial x^\nu}\right|\mathrm{d}^4x$$

$$\delta(\mathrm{d}^4x) = \mathrm{d}^4x' - \mathrm{d}^4x \approx \partial_\mu\delta x^\mu \mathrm{d}^4x,$$

since

$$\frac{\partial x'^\mu}{\partial x^\mu} = \frac{\partial(x^\mu + \delta x^\mu)}{\partial x^\nu} = g^\mu{}_\nu + \partial_\nu\delta x^\mu,$$

the property $\det A = e^{\mathrm{Tr}(\ln A)}$ yields

$$\det J(x) \underset{\delta x^\mu \to 0}{\approx} \exp\left[\mathrm{Tr}\,\partial_\nu\delta x^\mu\right] \sim 1 + \partial_\mu\delta x^\mu.$$

From this follows that the variation of the measure is

$$\delta(\mathrm{d}^4x) \approx \partial_\mu\delta x^\mu \mathrm{d}^4x.$$

[6]This is the definition of symmetry

So the action is stationary if and only if

$$0 = \Delta S = \int \mathrm{d}^4x \left[\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta\phi \right) + \underbrace{\partial_\mu \mathcal{L} \delta x^\mu + \mathcal{L} \partial_\mu \delta x^\mu}_{=\partial_\mu(\mathcal{L}\delta x^\mu)} \right] =$$
$$= \int \mathrm{d}^4x \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta\phi + \mathcal{L}\delta x^\mu \right) = 0$$

this happens if and only if

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta\phi + \mathcal{L}\delta x^\mu \right) = 0.$$

The function in parenthesis is *the Noether current*

$$J^\mu = \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta\phi + \mathcal{L}\delta x^\mu$$

and the found relation is its continuity equation, as

$$\partial_\mu J^\mu = \partial_0 J^0 + \partial_i J^i = \frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{j} = 0,$$

where $J^0 := \rho$, the density associated with the 3-current $\mathbf{j}$.

Integrating the continuity equation yields

$$0 = \int \mathrm{d}^3x \partial_\mu J^\mu = \int \mathrm{d}^3x \partial_0 J^0 + \underbrace{\int \mathrm{d}^3x \partial_i J^i}_{=0 \text{ by div thm}} = \int d^3x \frac{\partial J^0}{\partial t} = \frac{\mathrm{d}}{\mathrm{d}t} \int \mathrm{d}^3x J^0 := \frac{\mathrm{d}Q}{\mathrm{d}t} = 0$$

so the Noether charge which is conserved is

$$Q = \int d^3x J^0(x^\mu).$$

It is useful to rewrite the Noether current in terms of $\Delta\phi$: recalling that

$$\delta\phi = \Delta\phi - \partial_\nu \phi \delta x^\nu$$

the current becomes

$$J^\mu = \underbrace{\left(\mathcal{L} g^\mu{}_\nu - \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial_\nu \phi \right)}_{:=-T^\mu{}_\nu} \delta x^\nu + \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \Delta\phi$$

where $T^\mu{}_\nu$ is called the *stress energy tensor*; the sign is a convention that ensures $T^0{}_0 \geq 0$.

Lesson 9, 08/10

Observation The action is invariant under canonical transformations (because the E-L equation is), but the Lagrangian density is not: this produces an ambiguity in the Noether current J^μ, which is invariant under transformations in the form

$$J^\mu \to J'^\mu = J^\mu + \partial_\rho t^{\rho\mu}, \text{ with } t^{\rho\mu} = -t^{\mu\rho}$$

17.1 Transformation parameters

It is convenient to drop the dependency on the parameters of the transformation:

$$\delta x^\rho = \frac{\partial \delta x^\rho}{\partial \omega^A}\delta\omega^A$$
$$\delta\phi = \frac{\partial \delta\phi}{\partial \omega^A}\delta\omega^A$$

where ω^A are the parameters of the symmetry and A is a set of indices: in general the Noether current is in the form

$$J'^\mu = J^\mu{}_A \delta\omega^A$$

and, because the continuity equation holds $\forall\omega^A$,

$$\partial_\mu J^\mu{}_A \delta\omega^A = 0 \leftrightarrow \partial_\mu J^\mu{}_A = 0.$$

This form underlines the fact that the number of charges also depends on the indices of the parameters:

$$Q_A = \int \mathrm{d}^3x J^0_A(x^\mu) \rightarrow \frac{\mathrm{d}Q_A}{\mathrm{d}t} = 0.$$

17.2 Examples

- **Internal symmetries**
  Consider a field ϕ and its symmetry

  $$\phi \rightarrow \phi + \delta\phi, \quad \delta x^\mu = 0$$

  The Noether current is

  $$J^\mu = \frac{\partial \mathcal{L}}{\partial\partial_\mu\phi}\delta\phi$$

  and the charges are

  $$Q = \int \mathrm{d}^3x J^0$$

- **Space-time translations**
  Consider the transformation

  $$x^\mu \rightarrow x'^\mu = x^\mu + \varepsilon^\mu, \text{ with } \varepsilon^\mu << 1$$

  Since ϕ is a local field, $\Delta\phi = 0$ and

  $$\delta\phi = -\varepsilon^\mu\partial_\mu\phi$$

  The current is

  $$J^\mu = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi^\alpha}(-\varepsilon^\nu\partial_\nu\phi^\alpha) + \mathcal{L}\varepsilon^\mu = \varepsilon^\nu\left[\mathcal{L}g^\mu{}_\nu - \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi^\alpha}\partial_\nu\phi^\alpha\right] = -\varepsilon^\nu T^\mu{}_\nu$$

  as seen before, the continuity equation must hold $\forall\varepsilon^\nu$, so

  $$\partial_\mu T^\mu{}_\nu = 0.$$

  The Noether charges are

  $$Q^\mu := P^\mu = \int \mathrm{d}^3x T^0{}_\mu$$

  The 0-component is the total energy of the system, i.e. the Hamiltonian

  $$P^0 = H = \int \mathrm{d}^3x T^0{}_0 = \int \mathrm{d}^3x \mathcal{H}$$

where $\mathcal{H}$ is the Hamiltonian density.
This quantity can also be defined using the conjugated "coordinate" to ϕ

$$\pi(x^\mu) = \frac{\partial \mathcal{L}}{\partial \partial_0 \phi} = \frac{\partial \mathcal{L}}{\partial \dot{\phi}}$$

then

$$\mathcal{H} = \pi\dot{\phi} - \mathcal{L}$$

18 The action of a scalar field

The Lagrangian density of a scalar field is

$$\boxed{\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi(x^\mu))}$$

which includes a second derivative (since this is Lorentz-invariant) as a kinetic term and a potential term that depends only on ϕ.

The dimension of $\mathcal{L}$ turn out to be $[\mathcal{L}] = m^{-4} = kg^4$, so that of the field is $[\phi] = m^{-1} = kg$.

Using these dimensionality considerations, the simplest Lagrangian is the **Lagrangian of a non-interacting system**

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2.$$

From this, the E-L equation is the equation of motion for the field

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} &= 0\\ -m^2\phi - \partial_\mu(\partial^\mu\phi) &= 0\\ (\partial_\mu\partial^\mu + m^2)\phi(x^\mu) &= 0\end{aligned}$$

this is the K-G equation! This can also be interpreted as an eigenvalue equation for the Casimir operator $\Box = -P^2$, which implies

$$(-P^2 + m^2)\phi = 0 \leftrightarrow P^2 = m^2 :$$

this means that the dimensional parameter m in $\mathcal{L}$ is indeed the mass associated with the field.

18.1 Symmetries: translations

The Noether current for translations, the stress-energy tensor, in this case comes out to be

$$J_{\mu\nu} = T_{\mu\nu} = -g^{\mu\nu}\mathcal{L} + \partial_\mu\phi\partial_\nu\phi$$

and the charges, the momenta, are

$$P_\mu = \int \mathrm{d}^3x J_{\mu 0} = \int \mathrm{d}^3x(-g_{\mu 0}\mathcal{L} + \partial_\mu\phi\partial_0\phi).$$

The T_{00} component is

$$T_{00} = -\mathcal{L} + \partial_0\phi\partial_0\phi = \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}\partial_i\phi\partial^i\phi + \frac{1}{2}m^2\phi^2 := \mathcal{H}$$

18.2 Symmetries: Lorentz invariance

Here $\delta x^\mu = \varepsilon^{\mu\nu} x_\nu$ and, by definition, $\Delta\phi = 0$, so $\delta\phi = -\varepsilon^{\mu\nu} x_\nu \partial_\mu \phi$: the Noether current is

$$J^\mu = \left(\mathcal{L} g^\mu{}_\nu - \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial_\nu \phi \right) \delta x^\nu,$$

some boring calculations yield

$$J^\mu = -\varepsilon^{\nu\rho} \left(\partial^\mu \phi \partial_\nu \phi x_\rho - \mathcal{L} g^\mu{}_\nu x_\rho \right);$$

before eliminating the parameters $\varepsilon^{\nu\rho}$, it is necessary to antisymmetrize the parenthesis, the result is

$$J^\mu = -\frac{1}{2} \varepsilon^{\nu\rho} \underbrace{\left[\left(\partial^\mu \phi \partial_\nu \phi - \mathcal{L} g^\mu{}_\nu \right) x_\rho - \left(\partial^\mu \phi \partial_\rho \phi - \mathcal{L} g^\mu{}_\rho \right) \right]}_{:= J^\mu{}_{\nu\rho}}.$$

$$\partial_\mu J^\mu = 0 \leftrightarrow \partial_\mu J^\mu{}_{\nu\rho} = 0 \leftrightarrow \partial_\mu J^{\mu\nu\rho} = 0$$

So the parameter-independent Noether currents are $J^{\mu\nu\rho}$: they can be written in terms of the energy-momentum tensor

$$J^{\mu\nu\rho} = T^{\mu\nu} x^\rho - T^{\mu\rho} x^\nu.$$

It can be shown that the corresponding charges are the generators of Lorentz transformations in its infinite representation (?):

$$Q^{\nu\rho} = \int \mathrm{d}^3 x J^{0\nu\rho} = M^{\nu\rho} \to \frac{\mathrm{d}Q^{\nu\rho}}{\mathrm{d}t} = 0$$

so the angular momenta are conserved, since

$$L^i = \frac{1}{2} \varepsilon^{ijk} M_{jk},$$

and so are the boost generators

$$K^i = M^{0i}.$$

Lesson 10, 12/10

18.3 Charged scalar field

For the simple scalar field considered earlier, no internal symmetries can be found (?): to generalize the concept, one must consider a complex field, which has multiple components while retaining the fact that it is scalar: consider two independent real scalar fields ϕ_1, ϕ_2 and their linear combinations

$$\phi(x^\mu) = \frac{1}{\sqrt{2}} (\phi_1 + i\phi_2)$$
$$\phi^\dagger(x^\mu) = \frac{1}{\sqrt{2}} (\phi_1 - i\phi_2)$$

The simplest Lagrangian using this kind of field presents the simplest bilinear Lorentz invariance:

$$\mathcal{L} = \partial_\mu \phi \partial^\mu \phi^\dagger - m^2 \phi^\dagger \phi$$

The Euler-Lagrange equations turn out to be

$$\begin{cases} \left(\Box + m^2 \right) \phi(x^\mu) = 0 \\ \left(\Box + m^2 \right) \phi^\dagger(x^\mu) = 0 \end{cases}.$$

Consider the internal transformation that gives an arbitrary phase shift to the field

$$\phi \to \phi' = e^{i\alpha}\phi \to \phi' = \phi + i\alpha\phi$$
$$\phi \to \phi^{\dagger\prime} = e^{i\alpha}\phi \to \phi^{\dagger\prime} = \phi^\dagger + i\alpha\phi^\dagger$$

this is a symmetry with respect to the group $U(1)$.

The Noether current is

$$J^\mu = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\delta\phi + \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi^\dagger}\delta\phi^\dagger,$$

expanding the terms and factorizing the parameter α yields

$$J^\mu = \partial^\mu\phi^\dagger(i\phi) + \partial^\mu\phi(-i\phi^\dagger) = i\left(\phi^\dagger\partial_\mu\phi - \phi\partial_\mu\phi^\dagger\right).$$

Noether's theorem states that the quantity

$$Q = \int \mathrm{d}^3x J^0 = i\int \mathrm{d}^3x\left(\phi^\dagger\dot{\phi} - \phi\dot{\phi}^\dagger\right)$$

is conserved: in classical field theory, this does not seem like much, but with canonical quantization it will be clear that this Q is **the total charge of the system**.

19 Spinor Field

In this section the spinor field will be defined and its invariances will be studied.

19.1 Spinor representations of the Lorentz Group

As hinted before, there are 2 irreducible representations associated with spinor fields, both related to the $\mathfrak{su}(2)$ algebra:

$$\left(\frac{1}{2},0\right) \to \psi_L(x^\mu) = \begin{pmatrix}\psi_L^1(x^\mu)\\ \psi_L^2(x\mu)\end{pmatrix}$$

$$\left(0,\frac{1}{2}\right) \to \psi_R(x^\mu) = \begin{pmatrix}\psi_L^1(x^\mu)\\ \psi_L^2(x^\mu)\end{pmatrix}$$

Consider now the Lorentz transformations Λ_L and Λ_R, which are 2×2 matrices and are expected to be generated by the angular momenta J^i and boosts K^i operators, with the caveat, given by Wigner's theorem, that the K's do not admit unitary generators (??? CHECK):

$$J^i = \frac{1}{2}\sigma^i \to \exp\left[\frac{i}{2}\omega_i\sigma^i\right]$$
$$K^i = \mp i\frac{\sigma^i}{2} \to \exp\left[\mp\frac{1}{2}\eta_i\sigma^i\right]$$

Where the $\mp$ sign distinguishes

$$K_L^i = -i\frac{\sigma^i}{2},\ K_R^i = +i\frac{\sigma^i}{2}.$$

So the Lorentz transformations are

$$\Lambda_L = \exp\left[\frac{1}{2}\sigma^i\left(i\omega_i - \eta_i\right)\right]$$
$$\Lambda_R = \exp\left[\frac{1}{2}\sigma^i\left(i\omega_i + \eta_i\right)\right].$$

These transformation are related:

$$\Lambda_L^{-1} = \Lambda_R^\dagger,\ \sigma^2\Lambda_L\sigma^2,\ \Lambda_L^T\sigma^2\Lambda_L = \sigma^2,\ \Lambda_L^T = \sigma^2\Lambda_L^{-1}\sigma_2$$

and the relations hold switching R and L; these properties produce an interesting result

$$\sigma^2\psi_L^* \underset{\text{L.T.}}{\to} \sigma^2\Lambda_L^*\psi_L^* = \sigma^2\Lambda_L^*\sigma^2\sigma^2\psi_L^* = \Lambda_R\sigma^2\psi_L^*$$

which means that $\sigma^2\psi_L^*$ transforms like an element of the $\left(0, \frac{1}{2}\right)$ representation.

19.2 Invariants built from spinors

From QM1 spin composition, it is known that

$$\left(\frac{1}{2}, 0\right) \otimes \left(0, \frac{1}{2}\right) = (0,0) \oplus (1,0) :$$

consider two left spinors ψ_L, χ_L and the object

$$\chi^T\sigma^2\psi_L,$$

how does it transform?

$$\chi^T\sigma^2\psi_L \underset{\text{L.T.}}{\to} \chi_L^T\Lambda_L^T\sigma^2\Lambda_L\psi_L = \chi^T\sigma^2\psi_L$$

This combination of two spinors of the left representation is a scalar; consider the special case $\psi_L = \chi_L$:

$$\psi_L^T\sigma^2\psi_L = \begin{pmatrix}\psi_L^1 & \psi_L^2\end{pmatrix}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}\begin{pmatrix}\psi_L^1\\ \psi_L^2\end{pmatrix} = -i\psi_L^1\psi_L^2 + i\psi_L^2\psi_L^1$$

in order for this expression to be non-zero, the components of the spinors must be anticommuting numbers, called *Grassman Numbers*.

Considering $\chi_L = \sigma^2\psi_R^*$ and multiplying by i, yields

$$i(\sigma^2\psi_R^*)^T\sigma^2\psi_L = -i\psi_R^\dagger\psi_L$$

Equivalently, one can prove that another invariant is

$$\psi_L^\dagger\psi_R.$$

Another way to determine invariants uses the 4D extension of the Pauli matrices

$$\sigma^\mu = (\mathbb{I}, \sigma^i),$$
$$\bar{\sigma}^\mu = (\mathbb{I}, -\sigma^i)$$

to build 4-vectors like

$$\psi_L^\dagger\sigma^\mu\psi_L = (\psi_L^\dagger\psi_L, \psi_L^\dagger\sigma^i\psi_L)$$
$$\psi_R^\dagger\bar{\sigma}^\mu\psi_R = (\psi_R^\dagger\psi_R, -\psi_L^R\,\dagger\,\sigma^i\psi_R) :$$

one can use derivatives contracted with the index to create invariants.

19.3 Dirac Spinor

There is a problem with the 2 spinor representations, since the J's are pseudo-vectors and the K's are not, a parity transformation acts differently on the generators and on the Casimir operators: it interchanges the operators $N^2, N^{\dagger 2}$ and so the eigenvalues that label the representations

$$(n,m) \underset{\text{Parity}}{\longleftrightarrow} (m,n).$$

This means that one of these representations is not enough to describe a spin 1/2 field which is invariant under parity transformations (because reality is!).

The needed field is the **Dirac Spinor**, defined as

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} \psi_L^1 \\ \psi_L^2 \\ \psi_R^1 \\ \psi_R^2 \end{pmatrix}$$

19.4 Expanding the group

To unite the two spinors, a broader group needs to be used: recalling the definition of the sectors of the Lorentz Group, the needed group is

$$L^{\uparrow} = L_+^{\uparrow} \cup L_-^{\uparrow}$$

In this group, the sum of the two representations of the restricted Lorentz group is *an irreducible representation*: the generators are the 4 Dirac γ^μ matrices, which in the Weyl representations are

$$\gamma_0 = \begin{pmatrix} \mathbb{O} & \mathbb{I} \\ \mathbb{I} & \mathbb{O} \end{pmatrix}, \gamma^i = \begin{pmatrix} \mathbb{O} & \sigma^i \\ \sigma^i & \mathbb{O} \end{pmatrix}$$